In silico Neuronal Morphology Classification: A Systematic Review

Fábio Lobato[a,b,c,*], Jéssica Leite[a], Antonio Jacob Jr.[b], Roberto Santana[d]

[a] *Federal University of Western Pará, Institute of Engineering and Geosciences, Santarém, Pará, Brazil*
[b] *State University of Maranhão, Department of Computer Engineering, São Luís, Maranhão, Brazil*
[c] *University of São Paulo, Institute of Mathematics and Computer Sciences, São Carlos, São Paulo, Brazil*
[d] *University of the Basque Country, Department of Computer Science and Artificial Intelligence, Donostia/San Sebastián, Guipúzcoa, Spain*

List of Acronyms

Table 1: Glossary of acronyms and abbreviations used in the paper, with their respective descriptions.

Acronym	**Definition**
ANN	Artificial Neural Networks
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CRISP-DM	Cross-Industry Standard Process for Data Mining
DL	Deep Learning
DRNN	Deep Recurrent Neural Network
DST	Data Science Trajectories
DT	Decision Tree
EC	Exclusion Criteria
FCNN	Fully Convolutional Neural Network
fMRI	functional Magnetic Resonance Imaging
GNN	Graph Neural Network
GPT	Generative Pre-Trained Transformer
GPU	Graphics Processing Unit
IC	Inclusion Criteria
K-NN	K-Nearest Neighbors
LDA	Linear Discriminant Analysis
LSTM	Long Short-Term Memory
ML	Machine Learning
MLP	Multilayer Perceptron
PICOC	Population, Intervention, Comparison, Outcome, Context
QA	Quality Assessment
RF	Random Forest

*Corresponding author.
Email address: `fabio.lobato@ufopa.edu.br` (Fábio Lobato)

Acronym	Definition
RNN	Recurrent Neural Network
RQs	Research Questions
ResNet	Residual Network
SLR	Systematic Literature Review
SWC	File format for representing neuronal morphology, derived from the initials of its creators (Stockley, Wheal, and Cole)
SVM	Support Vector Machine
TMD	Topological Morphology Descriptor
TPU	Tensor Processor Unit
VGG	Visual Geometry Group
XGBoost	eXtreme Gradient Boosting